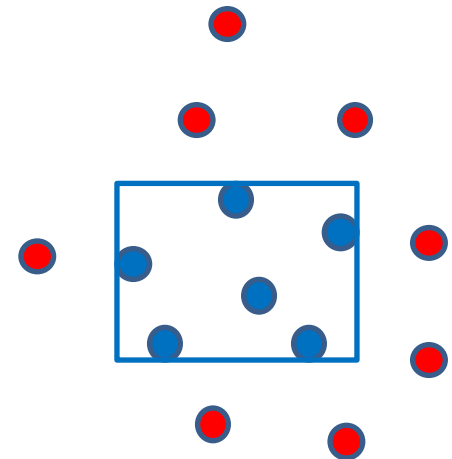
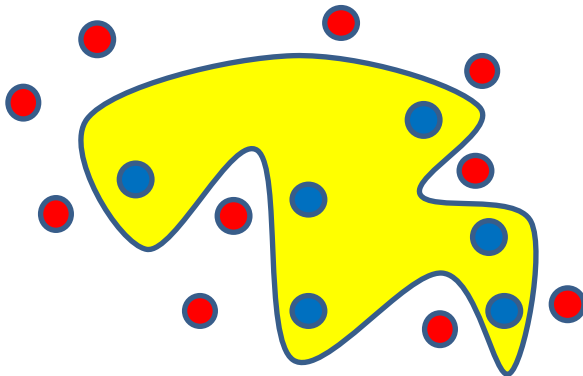


Shattering and VC-dimension

Lee-Ad Gottlieb

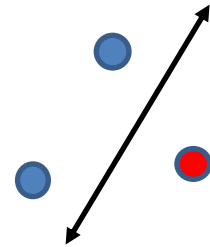
Learning

- Last week, we showed how to learn where our rules are
 - Infinite intervals
 - Infinite rectangles
 - Or any finite set of rules (even complex)

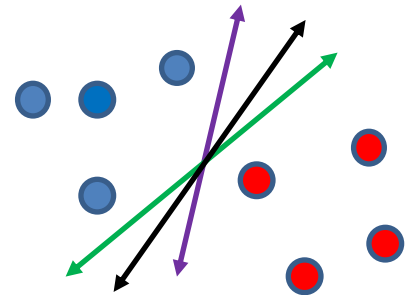


Learning and shattering

- What about an **infinite** set simple rules?
 - Lines, circles, etc.

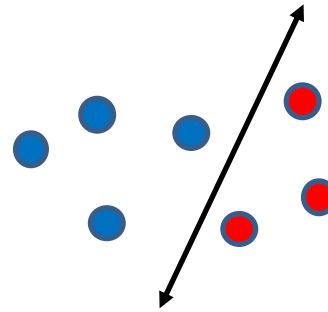
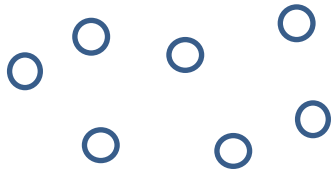


- Basic idea to handle this case:
 - Although one has an infinite number of lines, almost all of them do the same thing
 - Reduce to case of **finite** objects

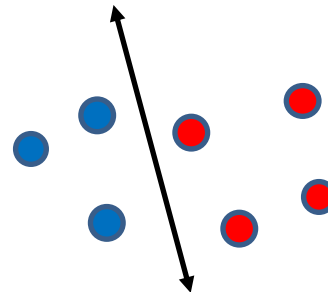
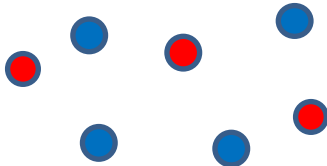


Learning and shattering

- We want to know how many different **behaviors** or **labelings** are possible.



- Then plug into analysis for finite point sets
 - Not all 2^7 behaviors can be achieved by a line

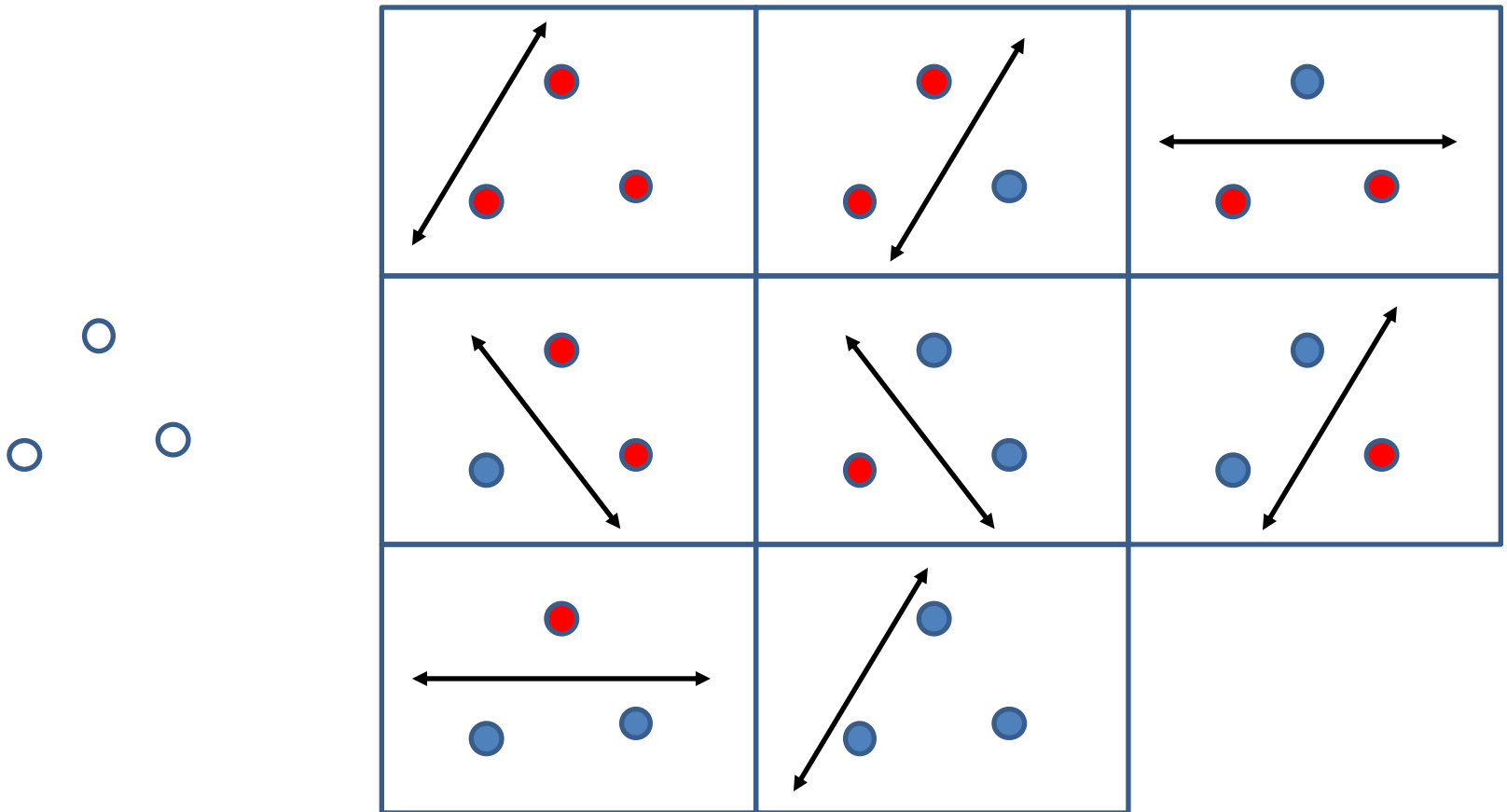


Learning and shattering

- Shattering:
 - Given a set of rules (hypothesis) H , and a set S of n points.
 - H **shatters** S if there is a rule in H achieving each possible labeling for S (2^n possible labelings)
- Similar, but not identical to the question of how many labelings are possible

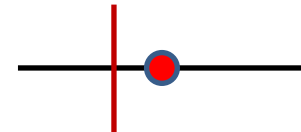
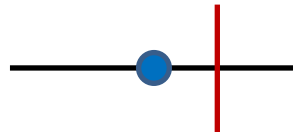
Shattering

- Example: the infinite set of (**bi-directional**) lines can shatter **some** (but not all) sets of 3 points in 2D

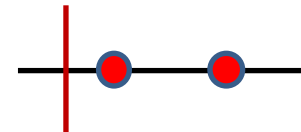
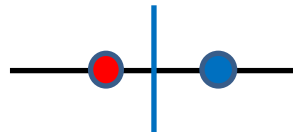
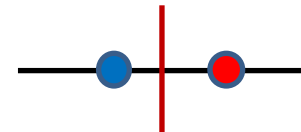
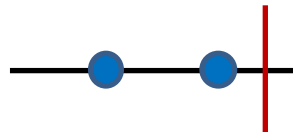


Shattering

- Example:
 - The **infinite** set of uni-directional intervals (right = red, left = blue) can shatter one point.
 - 2^1 labelings.

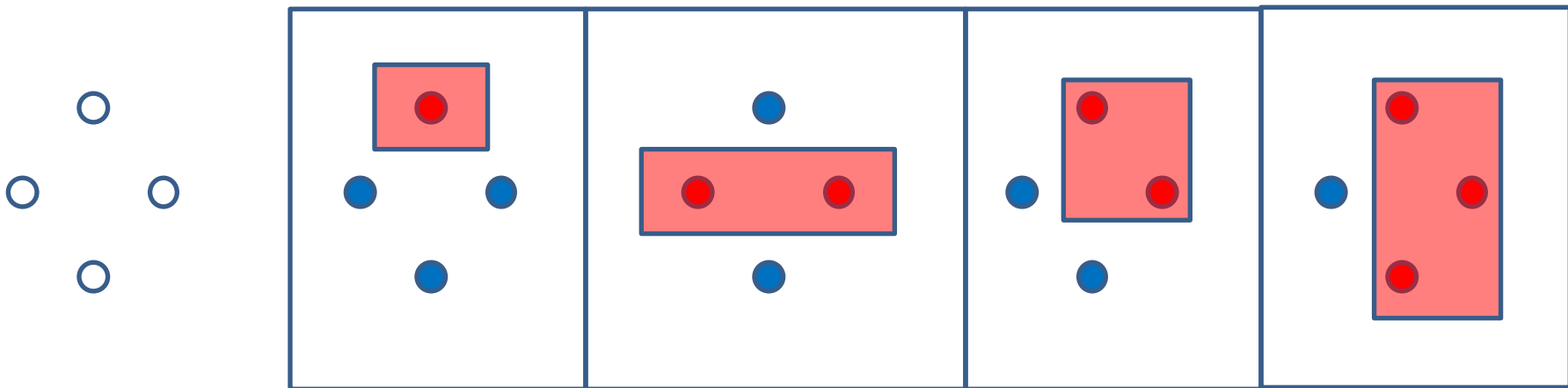


- The infinite set of bi-directional intervals can shatter two points.
- 2^2 labelings.



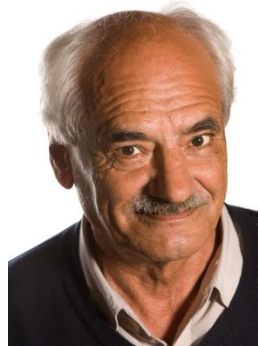
Shattering

- Example: The **infinite** set of uni-directional (inside = red, outside = blue) axis-aligned rectangles can shatter four points.
- 2^4 labelings. Four of them are drawn here:



VC-dimension

- Vapnik-Chervonenkis dimension



- The VC-dimension of a set H of hypotheses is the maximum number of points it can shatter.

VC-dimension

- The infinite set of **uni**-directional intervals (right = red, left = blue) has VC-dimension 1.
 - It can shatter one point.



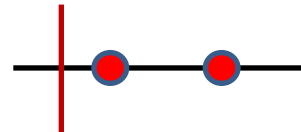
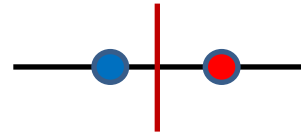
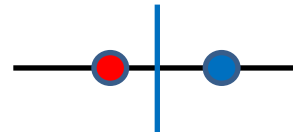
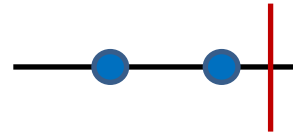
- It cannot shatter *any* set of two points, since it cannot make the left one red and the right one blue



VC-dimension

- The infinite set of bi-directional intervals has VC-dimension 2.

- It can shatter two points.

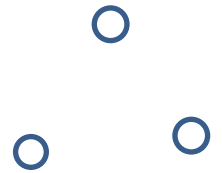


- It cannot shatter any set of three points, since it cannot give the end-point a different color than the middle point



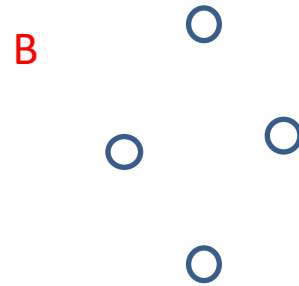
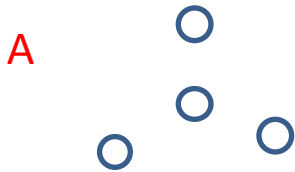
VC-dimension

- The infinite set of bi-directional **lines** has VC-dimension 3.
 - It can shatter *some* set of three points.
 - It cannot shatter *any* set of four points.
- Proof that lines cannot shatter any set of four points...



VC-dimension

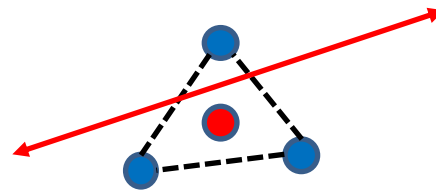
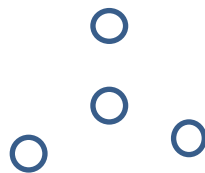
- Claim: There does not exist *any* set of four points which can be shattered by the infinite set of lines.



- Proof: Two possible cases
 - **A.** One point is contained in the convex hull of the others
 - **B.** The points are in general position.

VC-dimension

- **A.** One point is contained in the convex hull of the others
- A line cannot give the center point a different color than the outer points.
 - Such a line would need to separate the center point from an outer point →
 - So it goes through the triangle →
 - So it separates two outer points →
 - So it gives two outer points a different label



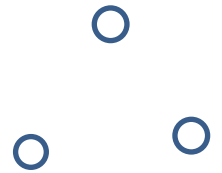
VC-dimension

- **B.** The points are in general position
- A line cannot give opposite endpoints different labels.
 - Since a line is linear, if two points are on one side of a line, so is any point in between.
 - There is a single point in between both pairs
 - It cannot be both on both the blue and red sides



VC-dimension

- The infinite set of bi-directional lines has VC-dimension 3.
 - It can shatter *some* set of three points.
 - It cannot shatter *any* set of four points.



There are sets of three points that a line can't shatter: three collinear points



VC-dimension

- The infinite set of uni-directional axis-parallel **rectangles** (inside = red, outside = blue) has VC-dimension 4.
 - It can shatter *some* set of 4 points
 - It cannot shatter *any* set of 5 points



- Proof: 2 cases
 - **A.** One point is in the convex hull of the others
 - **B.** The points are in general position

VC-dimension

- **A.** One point is contained in the convex hull of the others
- A rectangle cannot label the outer points red and the inner one green
 - The outer points must be inside the rectangle
 - A rectangle is a convex object →
 - The inner point must be inside as well.



VC-dimension

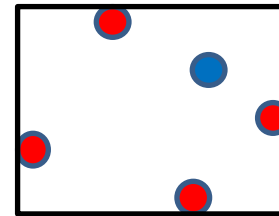
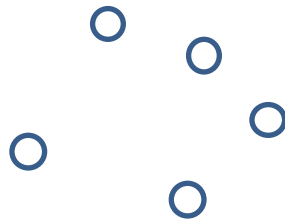
- **A.** One point is contained in the convex hull of the others
- A rectangle cannot label the outer points red and the inner one green
 - The outer points must be inside the rectangle
 - A rectangle is a **convex** object →
 - The inner point must be inside as well.

Proof works for
All convex objects



VC-dimension

- **B.** The points are in general position.
- Take the four points which the max, min x-values, and max, min y-values.
 - Any rectangle that contains them must contain the fifth point as well
 - Can't label the first four red and the last one green.



Sauer's lemma

- Sauer's lemma
 - Sauer-Shelah-Vapnik-Chervonenkis lemma.
- Given
 - Point-set S of size n
 - Rule set H with VC-dimension d
 - Let $\Pi(H, S)$ be all possible labelings that H assigns to S
- $\Pi(H, S) \leq \sum_{i=0}^d \binom{n}{i} = O(n^d)$
 - This could be much less than 2^n

Sauer's lemma

- Sauer's lemma
 - Sauer-Shelah-Vapnik-Chervonenkis lemma.

- Given

- Point-set S of size n
- Rule set H with VC-dimension d
- Let $\Pi(H, S)$ be the number of subsets of S that H assigns to S

Biggest term: $\binom{n}{i} = \frac{n!}{i!(n-i)!}$
 $< n \cdot (n-1) \dots (n-i+1)$
 $< n^i$
 Other terms: geometry series

H assigns to S

- $\Pi(H, S) \leq \sum_{i=0}^d \binom{n}{i} = O(n^d)$
 - This could be much less than 2^n

Sauer's lemma

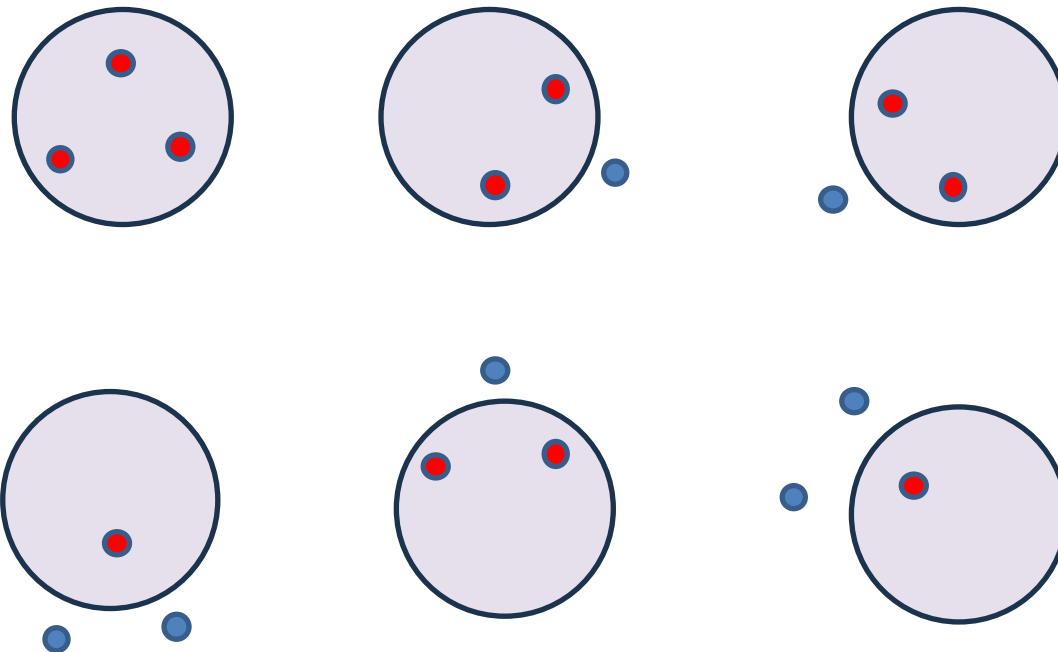
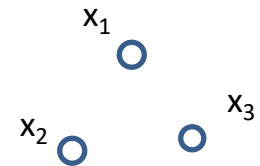
- Sauer's lemma
 - Sauer-Shelah-Vapnik-Chervonenkis lemma.
 - Given
 - Point-set S of size n
 - Rule set H with VC dimension d
 - Let $\Pi(H, S)$ be all possible assignments that H assigns to S
- More precisely,
 $\leq \left(\frac{en}{d}\right)^d$
- 
- $\Pi(H, S) \leq \sum_{i=0}^d \binom{n}{i} = O(n^d)$
 - This could be much less than 2^n

Preliminary

- $\binom{n}{i} = \binom{n-1}{i} + \binom{n-1}{i-1}$
- Intuition:
 - First term: didn't choose first element
 - Second term: Chose first element

Preliminary

- Some notation: labels **1**, **0**

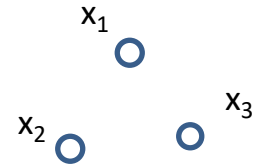


	x_1	x_2	x_3
h_1	0	0	0
h_2	0	0	1
h_3	0	1	0
h_4	0	1	1
h_5	1	0	0
h_7	1	1	0

H

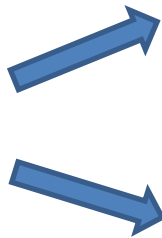
Sauer's lemma

- Proof: By induction.
 - Take all rules in H .
 - If two rules differ only on x_3 , put them in a different group.



H

	x_1	x_2	x_3
h_1	0	0	0
h_2	0	0	1
h_3	0	1	0
h_4	0	1	1
h_5	1	0	0
h_7	1	1	0



H_1

	x_1	x_2	x_3
h_1	0	0	0
h_3	0	1	0
h_5	1	0	0
h_7	1	1	0

H_2

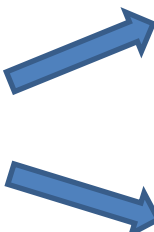
	x_1	x_2	x_3
h_2	0	0	1
h_4	0	1	1

Sauer's lemma

- The number of possible labelings for H_1, H_2 on S doesn't change if x_3 is erased: Let $S' = \{x_1, x_2\}$
 - $\Pi(H_1, S') = \Pi(H_1, S)$
 - $\Pi(H_2, S') = \Pi(H_2, S)$

H

	x_1	x_2	x_3
h_1	0	0	0
h_2	0	0	1
h_3	0	1	0
h_4	0	1	1
h_5	1	0	0
h_7	1	1	0



H_1

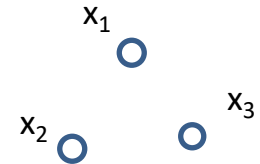
	x_1	x_2	x_3
h_1	0	0	0
h_3	0	1	0
h_5	1	0	0
h_7	1	1	0

H_2

	x_1	x_2	x_3
h_2	0	0	1
h_4	0	1	1

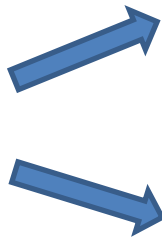
Sauer's lemma

- $\text{VC-dim}(H_1) \leq \text{VC-dim}(H) = d$



H

	x_1	x_2	x_3
h_1	0	0	0
h_2	0	0	1
h_3	0	1	0
h_4	0	1	1
h_5	1	0	0
h_7	1	1	0



H_1

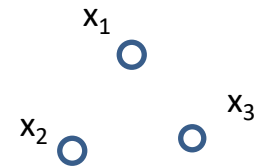
	x_1	x_2	x_3
h_1	0	0	0
h_3	0	1	0
h_5	1	0	0
h_7	1	1	0

H_2

	x_1	x_2	x_3
h_2	0	0	1
h_4	0	1	1

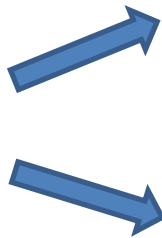
Sauer's lemma

- Take any set that H_2 can shatter
- H can shatter that set + x_3
- $VC\text{-dim}(H_2) \leq VC\text{-dim}(H) - 1 = d - 1$



H

	x_1	x_2	x_3
h_1	0	0	0
h_2	0	0	1
h_3	0	1	0
h_4	0	1	1
h_5	1	0	0
h_7	1	1	0



H_1

	x_1	x_2	x_3
h_1	0	0	0
h_3	0	1	0
h_5	1	0	0
h_7	1	1	0

H_2

	x_1	x_2	x_3
h_2	0	0	1
h_4	0	1	1

Sauer's lemma

- Proof by induction
 - Base case: For $n=d$ points, $\sum_{i=0}^d \binom{d}{i} = 2^d$ is simply all ways of labeling d points $\rightarrow 2^d$
 - Induction hypothesis: Assume it holds for $n-1$ points or less, and VC-dimension d or less
 - Prove it's correct for n points and VC-dimension d

Sauer's lemmas

- $\Pi(H, S) \leq \Pi(H_1, S) + \Pi(H_2, S)$
- $= \Pi(H_1, S') + \Pi(H_2, S')$
- $\leq \sum_{i=0}^d \binom{n-1}{i} + \sum_{i=0}^{d-1} \binom{n-1}{i}$
- $= \sum_{i=0}^d \binom{n-1}{i} + \sum_{i=1}^d \binom{n-1}{i-1}$
- $= \sum_{i=0}^d \binom{n-1}{i} + \sum_{i=0}^d \binom{n-1}{i-1}$
- $= \sum_{i=0}^d \binom{n}{i}$

Generalization bound

- We previously showed for finite set H and consistent classifiers

$$e(h) \leq \frac{1}{n} (\ln |H| + \ln \frac{1}{\delta})$$

- Plugging in $\left(\frac{en}{d}\right)^d$ instead of $|H|$ almost gives

$$e(h) \leq \frac{2}{n} (d \log_2 \frac{2en}{d} + \log_2 \frac{2}{\delta})$$

Generalization bounds

- We previously showed for finite set H

$$e(h) \leq \bar{e}(h) + \sqrt{\frac{\ln 2|H| + \ln \frac{1}{\delta}}{2n}}$$

- Plugging in $\left(\frac{en}{d}\right)^d$ instead of $|H|$ almost gives

$$e(h) \leq \bar{e}(h) + \sqrt{\frac{8d \ln \frac{2en}{d} + 8\ln \frac{4}{\delta}}{n}}$$